COMICCRAFT – AI COMIC STORY CREATOR USING GEMINI MODELS

6. Project Testing Phase

6.1 Introduction

The Project Testing Phase is performed to verify that all components of the ComicCraft application work correctly.

Testing helps identify errors, check the expected output, and improve the reliability of the application.

6.2 Types of Testing

The project includes the following types of testing:

- Unit Testing
- API Testing
- Integration Testing
- Functional Testing
- User Interface Testing
- Error Handling Testing

6.3 Unit Testing

Individual functions and modules are tested separately.

For example, utility functions and statistics-related functions are tested to ensure that they provide the expected results.

6.4 API Testing

The FastAPI endpoints are tested to verify that they respond correctly.

Important endpoints tested include:

- `/health`
- `/generate`
- `/generate-comic/json`
- `/test-image`

The health endpoint is used to check whether the application is running properly.

6.5 Gemini Testing

The Gemini integration is tested by providing sample story inputs.

The system is checked to verify whether it can generate:

- Story outline
- Narration
- Dialogues
- Image prompts

6.6 Image Generation Testing

The image generation functionality is tested using sample prompts.

The generated images are checked to ensure that they are successfully created and used in the comic panels.

6.7 User Interface Testing

The web interface is tested to verify:

- Input fields
- Buttons
- Form submission
- Comic preview
- Generated content
- PDF export

6.8 Error Handling Testing

The application is tested with incomplete or incorrect inputs.

Configuration errors, API errors, and image-generation errors are also checked to ensure that the application handles problems properly.

6.9 Testing Outcome

Testing helps identify and correct errors before the final demonstration.

The final application should successfully accept user input, generate comic content, display comic panels, and provide PDF export functionality.